

Air Column Resonance: Open vs Closed Pipes

Air Column Resonance · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

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Simulation: <https://ophysics.com/w10.html>

Curriculum links: waves

Learning intentions

- I can find resonant lengths for open and closed air columns at a fixed frequency.
- I can relate pipe length to wavelength for each type of pipe.
- I can calculate the speed of sound from a measured resonance.

Getting started

Open the simulation and select the closed-pipe setting. Set a fixed driving frequency, then slowly change the pipe length until you find the first point of resonance.

Tasks

1. For the closed pipe, find the first three resonant lengths at your chosen frequency and record them.

Resonance number

Pipe length

2. Compare the ratios of your three lengths from Task 1 to the expected ratio of 1 : 3 : 5 for a closed pipe.

3. Switch to the open pipe setting and find its first three resonant lengths at the same frequency.

Resonance number

Pipe length

4. Using your first resonant length and the known frequency, calculate the speed of sound implied by the simulation.

Working:

Extension (optional)

Explain why a closed pipe only produces odd-numbered harmonics, while an open pipe produces all harmonics.

